College of Technology Galle

Course: Full Stack Software Development — Object Oriented Programming — Module K72T004M03 — NVQ Level 05

SEMESTER MODEL PAPER 10

Focus: Capstone-style design, polymorphism mastery, critique & extension

Difficulty: Hard

Duration: 3 Hours **Total Marks:** 100 **Questions:** 7 structured questions (answer any **5**) **Each question:** 20 marks

Instructions to Candidates

1. Answer any **five (5)** questions out of the seven provided.
2. Each question carries **20 marks**. Maximum mark for this paper is **100**.
3. Write answers in the answer booklet. Do **not** write answers on this question paper.
4. Where Java code is required, write clear, syntactically correct Java as taught in this module (VS Code / JDK).
5. Show workings / short explanations where asked. Manage your time (about 35 minutes per question).
6. Mobile phones and programmable devices are not allowed.

Question 1 [20 Marks]

Capstone design brief.

a. You must design a console "Campus Lost & Found". Identify at least four classes and their relationships (inherit/associate/compose). **(8 marks)**

b. Write full Java for an abstract `Item` and two subclasses with one distinct behaviour each. **(8 marks)**

c. Describe how a polymorphic list of items would be displayed. **(4 marks)**

Question 2 [20 Marks]

Contract-first with interfaces.

a. Define `Searchable` and `Auditable` interfaces. **(4 marks)**

b. Implement a `BorrowRecord` that supports both; show a method that accepts `Searchable` only. **(10 marks)**

c. Explain compile-time vs runtime roles of the interface type. **(6 marks)**

Question 3 [20 Marks]

Complex override rules.

a. Write a superclass method that returns `Animal` and a subclass override that returns `Dog` (covariant return) - if not taught, explain the safer same-return override pattern instead and implement that. **(8 marks)**

b. Demonstrate calling the override through a superclass reference. **(6 marks)**

c. Discuss whether making the superclass method `final` would help or hurt extensibility here. **(6 marks)**

Question 4 [20 Marks]

Integrity with composition.

a. Model an `Order` composed of `OrderLine` parts; totals derived from lines. **(8 marks)**

b. Prevent empty orders and negative quantities with exceptions. **(6 marks)**

c. Explain cascading lifetime: what happens conceptually when the order is discarded. **(6 marks)**

Question 5 [20 Marks]

Performance & memory awareness.

a. Explain how creating thousands of temporary `String` objects in a loop differs from updating a numeric counter field. **(6 marks)**

b. Given a library catalogue array, discuss reference cost vs copying objects when "sorting references". **(8 marks)**

c. How does understanding stack/heap help debug a stubborn NullPointerException? **(6 marks)**

Question 6 [20 Marks]

Extension without breaking clients.

a. An existing `Printer` class is used widely. You must add colour printing variants. Propose an inheritance or interface-based extension that minimises breakage. **(8 marks)**

b. Provide class/interface sketches and a short client migration note. **(8 marks)**

c. What regression tests would you run from `main` after the change? **(4 marks)**

Question 7 [20 Marks]

Examiner critique.

a. A student submits a single 400-line `Main` with no classes. Mark scheme style: award/deduct conceptual marks across OOP pillars - write a structured examiner comment (strengths/weaknesses/advice). **(10 marks)**

b. Outline a refactor plan into at least five types (classes/interfaces) with responsibilities. **(6 marks)**

c. Write the method signatures only for the public API of the two most important types. **(4 marks)**

******* END OF QUESTION PAPER *******